

EN 612: Structure - Property correlation in materials for Energy Applications

Course Code	EN 612
Course Name	Structure - Property correlation in materials for Energy Applications
L-T-P-C	2-0-2-3
Prerequisites	IC241 (Materials Science for Engineers)/ Instructor Contest
Intended for	UG/M. Tech./ MS/ PhD
Distribution	Compulsory for M. Tech. in Energy Engineering with specialization in Materials, and Elective for others
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**
Structure of Materials – Crystal structure, micro structure and macrostructure; Determination of crystal structure by X-ray diffraction and microstructure by optical, scanning and electron microscopy, selected area diffraction; Phase diagram of binary alloys and distribution of phases in microstructure of cast and wrought alloys [10 Lectures]
- **Module II**
Ni-based super-alloys – Austenitic Ni-Cr binary alloys, Ni-based superalloys based on γ phase containing, nickel, cobalt, iron, chromium and molybdenum, Coherent and incoherent precipitates, γ (Ni₃Al based L12 structure) precipitates, nickel – iron superalloys (IN718 and IN706) hardened by γ'' (Ni₃Nb based D022 structure), basal plane coherency, Grain growth and pinning of grain boundaries, Carbides and borides in superalloys; refractory elements and hard intermetallic TCP phases, Influence of microstructure on strength, ductility and creep life. Iron and cobalt based superalloys. [10 Lectures]
- **Module III**
Conductors and semiconductor devices, solar photovoltaic materials, organic photovoltaic materials. [8 Lectures]

Laboratory Modules

1. Determination of crystal structure by X-ray diffraction in a diffractometer
2. Preparation of specimen and observation of microstructure in single phase alloy under optical microscope
3. Microstructure of cast and wrought alloys under optical microscope
4. Microstructure of iron based alloys in hydroturbines under optical and electron microscope
5. Microstructure of Ni-based superalloys in steam and gas turbines under optical and electron microscope along with SAD of precipitates.
6. Microstructure of alloys after service – erosion and cavitation in hydroturbines; creep cavitation in high temperature turbines
7. Microstructure of silicon wafers under optical and electron microscope

References

1. Robert E Reed-Hill and Reza Abbaschian, **Physical Metallurgy Principles**, Thomson, 2003 reprint.
2. Chester T Sims, William C Hagel, **Superalloys II**, Wiley-Interscience, 1987
3. R. E. Hummel, **Electronic Properties of Materials**, 4th Edition, Springer, 2011.